

APR.-17/B.E./Insem.-3

B.E. (Civil)

ADVANCED STRUCTURAL DESIGN

(Semester - II) (2012 Pattern) (Elective - III)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4 and Q.5 or Q.6.
- 2) Figures to the right indicate full marks.
- 3) All relevant IS Codes and Steel Table are allowed in the examination.
- 4) If necessary, assume suitable data and indicate clearly.
- 5) Use of electronic pocket calculator is allowed.

Q1) Determine the allowable deflection for a beam of effective span 4 m. The beam section is fabricated using two channels with bent lips connected along their webs. The channels are 200 mm × 80 mm with 25 mm bent lips. The thickness of the plate is 2.5 mm. Consider the yield stress in steel as 235 N/mm². [10]

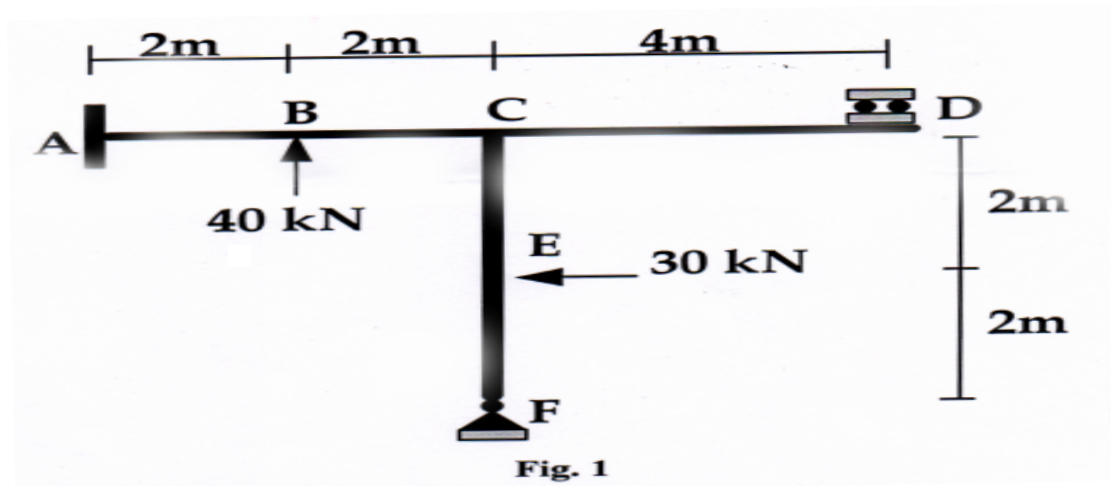
OR

- Q2)** a) Explain effective width used in the design of compression elements. [5]
 b) Explain stiffened and un-stiffened elements with neat sketches. [5]

Q3) Explain the theorems of plasticity used in the analysis of steel frames. [10]

OR

Q4) Determine the collapse moment for the frame shown in Fig. 1 and plot the bending moment diagram. [10]



P.T.O.

Q5) Calculate the overturning moments for a self-supporting chimney which has a total height of 50 m above the foundation. The chimney has uniform diameter of 2 m and is lined with 100 mm thick brick lining. The location of the chimney is on a site having a basic wind speed of 32 m/s. The topography of the site is flat. [10]

OR

Q6) Explain with a neat sketch how the preliminary dimensions for a steel chimney are worked out. [10]

